BHAKTI LAMBATE

HARDWARE ENGINEER | SILICON ARCHITECTURE & RTL DESIGN

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SUMMARY

Driven Electronics & VLSI Engineering student (Class of 2027) with a strong technical foundation in Digital Logic Design, Computer Architecture, and RTL Development. Proficient in Verilog and SystemVerilog, with a track record of designing optimized arithmetic units and pipelined processors on FPGA platforms (Xilinx/Spartan). Skilled in Static Timing Analysis (STA), PPA (Power, Performance, Area) optimization, and automation using Python and C++. Possess a first-principles approach to hardware-software co-design and a deep interest in Parallel Computing and AI-accelerated silicon. Eager to leverage analytical problem-solving skills and 'first-time-right' design methodologies to contribute to cutting-edge hardware cycles at a global semiconductor leader.

SKILLS & TECHNOLOGIES

Design & RTL	Verilog, SystemVerilog, RTL Design & Microarchitecture, Clock Domain Crossing (CDC), SoC Design
Verification	UVM (Universal Verification Methodology), Simulation (ModelSim), Testbench Development, Formal Verification
Software & Scripting	Python, Perl, TCL, C & C++, Linux Shell Scripting (Bash), Data Structures & Algorithms
Tools	Xilinx Vivado, Synopsys Design Compiler (Synthesis), MATLAB, PrimeTime (STA), Git, CUDA, Intel Quartus
Soft Skills	Leadership, Problem-Solving, Teamwork & Communication, Analytical skills, Attention to Detail, Work Ethic

PROJECTS

FPGA Digital Signal Processor (DSP) Implementation | Xilinx Spartan-7 | March 2026- April 2026

- Designed and implemented a high-performance DSP pipeline on a Xilinx Spartan-7 using Verilog HDL, managing the full RTL-to-bitstream flow in Vivado and optimizing logic by mapping operations to dedicated DSP48E1 slices.
- Achieved timing closure via Static Timing Analysis (STA) and validated real-time hardware functionality using Integrated Logic Analyzer (ILA) cores and self-checking testbenches to balance LUT utilization and speed.

FPGA Adder Architecture Design | Xilinx Spartan-7 | Jan 2026- Feb 2026

- Designed and implemented 4-bit Ripple Carry (RCA) and Carry Lookahead (CLA) adder architectures on a Xilinx Spartan-7 FPGA using Verilog HDL. Managed the complete RTL-to-bitstream flow within Xilinx Vivado, focusing on optimizing computational speed by reducing propagation delay through CLA logic.
- Conducted thorough functional verification via testbenches and validated the final design on hardware using physical I/O peripherals, successfully analyzing the trade-offs between timing performance and LUT utilization.

ACHIEVEMENTS

- **Smart India Hackathon 2025 Finalist:** Recognized as a Finalist in the nationwide Smart India Hackathon 2025, showcasing expertise in rapid development and innovative technical solutions.
- **Fellowship- International VLSI Design Conference 2026:** Engaged in technical sessions on semiconductor design and RISC-V architectures. Collaborated with industry experts and EDA leaders to explore emerging trends in the global semiconductor ecosystem.

EXPERIENCE

- **System Testing Intern | National Test House | Government of India** — **Dec 2025- Jan 2026**
  Ministry of Consumer Affairs, Food & Public Distribution
  - Drove the end-to-end quality assurance lifecycle and spearheaded the rigorous verification and validation of a suite of mission-critical platforms, including a sophisticated Laboratory Information Management System (LIMS) and an enterprise-grade Procurement Platform.
  - Mandated and ensured absolute compliance with stringent central government regulatory standards, establishing and upholding rigorous quality assurance (QA) protocols that exceeded minimum mandated requirements.
- **MERN Stack Development Trainee| Contentstack** — **Feb 2025 - July 2025**
  EmpowerHer Training Program
  - Pioneered a MERN stack platform (MediLink) for holistic medical appointment and digital health record orchestration, gaining practical experience in data management and accessibility.
  - Features a fortified web interface with intelligent OCR integration, significantly enhancing data management and accessibility, showcasing the practical application of technology to improve data handling.

EDUCATION

- **Bachelor of Engineering - Electronics Engineering (VLSI Design & Technology)** — **July 2023- May 2027**
  Honors- Artificial Intelligence & Machine Learning
  Current CGPA- 8.68
- **12th Grade- HSC (Computer Science)** — **June 2021- April 2023**
  Percentage- 60%
- **10th Grade- SSC Maharashtra Board** — **June 2011- March 2021**
  Percentage- 86.40%